

LINKEDLIST

Q4)

Question:

DS	Session	Linked List	Question Information	Level 1	Challenge 34
Question description Problem Description: For Engineering Second -year students, the instructor is organising a surprise test. He's given you a new node is added before the given node of the given Linked List. For example if the given Linked List is 5->10->15->20->25 and we add an item 30 before node 15, then the Linked List becomes 5->10->30->15->20->25. Since a Linked List is typically represented by the head of it, we have to traverse the list till node and then insert the node. Constraints: 1 < arr <100 Input Format First line contains the number of datas- N. Second line contains N integers(the given linked list). Third line contains the node P before which the node to be inserted. Fourth line contain the node X to be inserted.					

PROGRAM:

```
1  #include <stdio.h>
2  #include <stdlib.h>
3
4  struct Node {
5      int data;
6      struct Node *next;
7  };
8
9  struct Node* createNode(int data) {
10     struct Node *newNode = (struct Node*)malloc(sizeof(struct Node));
11     newNode->data = data;
12     newNode->next = NULL;
13     return newNode;
14 }
15
16 struct Node* insertEnd(struct Node *head, int data) {
17     struct Node *newNode = createNode(data);
18
19     if (head == NULL)
20         return newNode;
21
22     struct Node *temp = head;
23
24     while (temp->next != NULL)
25         temp = temp->next;
26
27     temp->next = newNode;
28
29     return head;
30 }
31
32 struct Node* insertBefore(struct Node *head, int P, int X, int *found) {
33     struct Node *newNode;
34
35     if (head == NULL)
36         return head;
37
38     if (head->data == P) {
39         newNode = createNode(X);
40         newNode->next = head;
41         *found = 1;
42         return newNode;
43     }
44
45     struct Node *temp = head;
46
47     while (temp->next != NULL) {
48         if (temp->next->data == P) {
49             newNode = createNode(X);
50             newNode->next = temp->next;
51             temp->next = newNode;
52             *found = 1;
53             return head;
54         }
55
56         temp = temp->next;
57     }
58
59     return head;
60 }
```

```
int main() {
    int N, P, X, data;
    int found = 0;
    struct Node *head = NULL;

    scanf("%d", &N);

    for (int i = 0; i < N; i++) {
        scanf("%d", &data);
        head = insertEnd(head, data);
    }

    scanf("%d", &P);
    scanf("%d", &X);

    head = insertBefore(head, P, X, &found);

    if (!found)
        printf("Node not found!\n");

    display(head);

    return 0;
}
```

OUTPUT:

TEST CASE 1:

```
4
115 210 116 110
116
17
Linked List:->115->210->17->116->110
```

TEST CASE 2:

```
4
15 20 16 10
25
17
Node not found!
Linked List:->15->20->16->10
```